VoltGuard-ECM: A DMS Integration Architecture for AC-Audited Voltage-Risk Screening in High-DER Distribution Networks

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Abstract

High-DER distribution operation requires distribution management systems (DMS) to triage many forecasted feeder states before committing expensive AC power flow, OPF, or MPC studies. This paper addresses a system-integration question rather than a new learning-method question: how should a calibrated voltage-risk screen be embedded into a DMS so that release, watch, and corrective-audit decisions remain auditable? VoltGuard-ECM specifies an integration architecture, input data contract, queue semantics, exception rules, algorithmic decision logic, rolling drift audit, and AC-audit handoff for a screening layer placed between forecast ingestion and trusted corrective tools. The paper deliberately does not reuse the core prediction, calibration, and release metrics of the companion method study. Its contribution is the engineering envelope that makes any calibrated screen deployable: input validity checks, queue labels, operator override logic, rolling drift audit, and mandatory fallback to AC analysis under stale telemetry, unseen topology, or out-of-envelope DER forecasts. A seven-day Python prototype processes 1008 simulated operating cycles and demonstrates release/watch/corrective-audit queue assignment, fallback routing, operator override logging, and drift-alarm records. VoltGuard-ECM is therefore a DMS-facing governance layer for voltage-risk screening, not a claim that a screen replaces AC feasibility analysis or optimal control.

Keywords

Distribution operation; Voltage-risk screening; DMS integration; Queue semantics; DER operation; AC audit; Fallback rules

1. Engineering Motivation

Distribution control rooms increasingly receive batches of PV, EV, storage, and flexible-load forecasts. A conservative workflow sends every forecasted state to AC power flow or corrective optimization. That workflow is robust but can be inefficient when many scenarios are clearly far from voltage limits. A screening layer is attractive only if it can be integrated without weakening operator accountability.

This paper asks a different question from a method paper. It does not ask which residual learner or conformal grouping gives the best interval. It asks: what DMS contract is needed before a voltage-risk screen can safely influence release, watch, and audit queues?

The answer is an architecture. VoltGuard-ECM treats the screen as a governed front end. It receives forecast and topology objects, returns auditable queue records, and triggers fallback rules whenever the inputs or residual behavior fall outside the calibrated envelope.

This is the DMS companion paper to the VoltGuard screening-method study and the LinDistFlow-global-quantile baseline note. The method paper defines how calibrated intervals are estimated; the baseline note defines a transparent reference screen; this paper defines the operational contract required before any such screen may influence DMS queues.

2. Engineering Context

Distribution operation already relies on AC power flow, state estimation, operator rules, voltage margins, OPF, and corrective control. These tools remain the authority for feasibility. A screening layer can reduce workload only if it is subordinate to those tools and if every release decision remains traceable.

The integration challenge is therefore not only computational. It is also a data-governance problem. The screen must know which feeder model was used, which topology state is active, which forecasts are fresh, which calibration protocol applies, and when the result should be ignored. Without this contract, a strong offline predictor can become unsafe in deployment because the DMS cannot tell whether the current state is comparable to calibration.

2.1. Acronym Table

Acronym	Meaning
AC	alternating-current power flow model
AMI	advanced metering infrastructure
CIM	common information model
DER	distributed energy resource
DMS	distribution management system

Acronym	Meaning
EV	electric vehicle
IEC	International Electrotechnical Commission
MPC	model predictive control
OPF	optimal power flow
PV	photovoltaic generation
SCADA	supervisory control and data acquisition

2.2. DMS Integration Standards and Protocols

The architecture should be mapped to existing utility data standards rather than implemented as an isolated ML service. IEC 61970/61968 CIM objects [27,28] can carry network models, equipment identifiers, measurements, switch states, and asset metadata. IEEE 2030.5 [29] can represent DER communication and control interfaces in customer-side or aggregator-connected settings. OpenFMB-style [30] field-message patterns can support distributed message exchange when feeder devices, edge applications, and control-room services share a common operational model. VoltGuard-ECM does not prescribe one vendor stack; it specifies the minimum screening fields that must survive translation into those standards: model version, topology id, forecast timestamp, calibration protocol, queue label, reason code, operator override, and downstream AC outcome.

2.3. Generalization to Other Screening Methods

The architecture is designed for VoltGuard but is not VoltGuard-specific. A LinDistFlow-global-quantile screen, a sensitivity-factor screen, a quantile regression model, or a future OPF-assisted classifier can plug into the DMS contract if it exposes four outputs: a scenario risk score, bus-level risk flags or voltage intervals, a calibration/envelope identifier, and a reason code when the screen declines to release a scenario. The DMS should treat the screen as an interchangeable module and should validate each module with the same queue, fallback, audit, and drift-monitoring tests.

2.4. Related Integration Work and Positioning

Prior work on DMS and energy-management integration can be grouped into four technical lines. First, model-exchange standards such as CIM, IEEE 2030.5, and OpenFMB define how grid assets, measurements, DER messages, and field devices should be represented, but they do not specify how an ML-based voltage screen should decide release, watch, and audit queues. Second, conventional DMS analytics rely on state estimation, AC power flow, OPF, and operator voltage margins. These tools remain authoritative, yet they can be computationally expensive when every forecasted DER scenario is audited with full AC analysis. Third, edge-computing and federated-learning studies for distribution systems address where analytics may run and how data can be shared, but they often leave queue semantics, reason codes, and operator overrides underspecified. Fourth, ML voltage-prediction papers report offline accuracy, whereas deployment requires input-validity checks, model-version binding, and post-decision audit records.

VoltGuard-ECM is positioned at this integration boundary. It does not replace the standards or the AC tools; it defines the missing operational envelope that allows a calibrated screen to become a traceable DMS object.

Integration line	Main contribution	Gap for voltage-risk screening	VoltGuard-ECM role
CIM / IEEE 2030.5 / OpenFMB standards	interoperable grid and DER data objects	no queue or fallback policy for screening outputs	maps screen records to auditable fields
AC PF / OPF / MPC in DMS	trusted feasibility and corrective decisions	expensive for high-volume forecast triage	preserves AC audit as backend authority
Edge and federated grid analytics	deployment location and data-sharing pattern	weak operator-facing semantics	adds release/watch/audit reason codes
ML voltage prediction papers	offline point or interval estimates	limited DMS governance and drift logging	wraps calibrated screens in a DMS contract

3. System Integration Architecture

VoltGuard-ECM contains five DMS-facing interfaces.

First, the forecast interface receives load, PV, EV, storage, and flexible-load forecasts or pseudo-measurements with timestamps and provenance. Each forecast record must identify the operating horizon, data source, freshness, and quality flags.

Second, the network-model interface receives feeder topology, branch impedances, slack-bus voltage, switch states, voltage limits, and DER connection metadata. The screening layer must bind every result to a specific model version and topology identifier.

Third, the screening interface computes voltage-risk intervals or point-risk flags using the active screening module selected by the operator. The architecture is method-agnostic: the module can be a LinDistFlow baseline, residual learner, conformal interval estimator, or another calibrated screen.

Fourth, the queue interface maps scenario-level outputs to release, watch, and corrective-audit queues. The release queue is only advisory: it means that the scenario does not require immediate AC audit under the current envelope. The watch queue preserves cases close to limits or affected by mild input uncertainty. The corrective-audit queue sends cases to AC power flow, OPF, MPC, or operator review.

Fifth, the audit interface records the decision path. Each record stores the scenario identifier, model version, calibration protocol, input-validity checks, risk score, queue label, operator override if present, and downstream AC outcome once available.

The complete workflow is: forecast ingestion; CIM/topology binding; input validity checks; screening-module execution; fallback-rule evaluation; queue assignment; operator review or override; AC power-flow/OPF/MPC handoff for audited cases; downstream outcome logging; and rolling residual/drift update. The graphical workflow in Figure 1 gives the high-level flow, while the algorithms below specify the executable decision logic.

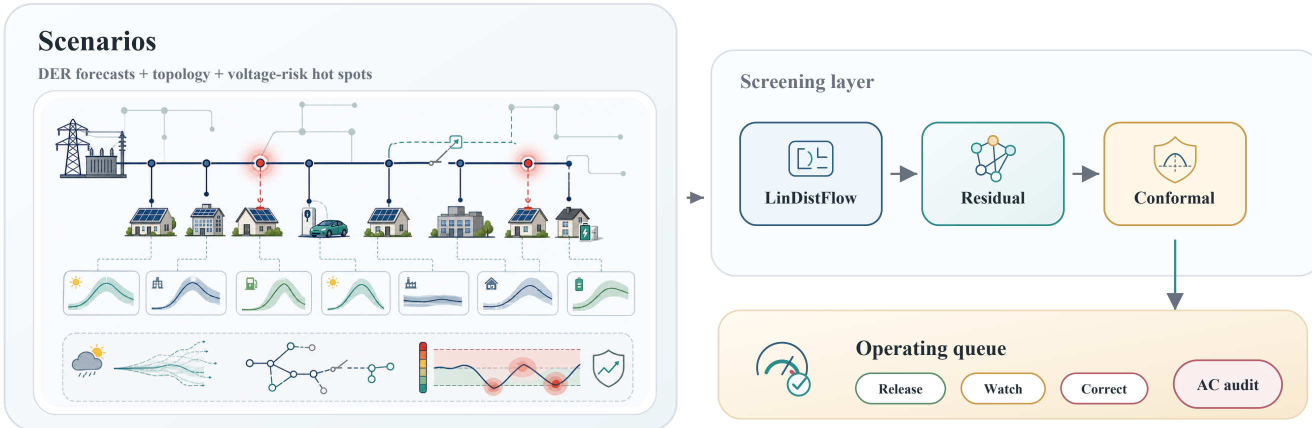

Figure 1: Graphical abstract and VoltGuard screening workflow. The figure summarizes the active-feeder inputs, LinDistFlow physical backbone, topology-aware residual correction, conformal voltage-risk intervals, and downstream AC-audited corrective optimization interface.

4. Stakeholder Requirements

Operators need a screen they can interrogate. Every release, watch, or audit record should show the active topology, the voltage-limit margin, the reason code, and whether the result came from a calibrated family or a fallback rule. The system should minimize nuisance alarms, but operator trust depends more on clear escalation rules than on a single offline accuracy number.

Utility IT teams need cybersecurity, access control, logging, and data lineage. Screening modules should run with least-privilege access, receive signed model artifacts, write immutable audit records, and avoid storing unnecessary customer-level data outside approved systems. Model files, calibration sets, and queue records should have retention policies consistent with the utility's operational data-governance program.

Regulators and compliance auditors need accountability. The DMS must be able to reconstruct why a scenario was released or sent to AC audit, which model version was active, whether an operator overrode the queue label, and what the downstream AC result showed. Black-box predictions are therefore insufficient; the screening layer must produce traceable decisions.

5. Queue Semantics

5.1. Release Queue

A released scenario is not certified feasible. It is a scenario that the screening layer considers low priority for immediate AC audit under the active calibration envelope. Release records should include the interval margin to voltage limits, the active topology identifier, and the input-validity status. Operators may still sample released scenarios for periodic AC audit to detect silent drift.

5.2. Watch Queue

The watch queue is the buffer between release and corrective audit. Scenarios enter this queue when intervals are near voltage limits, forecast quality is degraded, or recent residual monitoring suggests mild distribution shift. The DMS can route watch scenarios to delayed AC audit, operator review, or additional measurement collection.

5.3. Corrective-Audit Queue

The corrective-audit queue is mandatory when intervals intersect voltage limits, input checks fail, topology is unseen, or telemetry is stale. The screen does not prescribe the final action. It supplies the scenario, suspected buses, direction of voltage risk, and ranking metadata to the downstream AC tool.

6. Fallback and Drift Governance

VoltGuard-ECM uses conservative fallback rules because calibrated screens are protocol-dependent. The following conditions bypass release and trigger AC audit or operator review:

- feeder topology is missing, stale, or absent from the calibration envelope;
- forecast timestamps exceed the DMS freshness limit;
- PV, EV, storage, or load features lie outside calibrated ranges;
- online residual monitors exceed a rolling threshold;
- protection events, storm conditions, switching operations, or known data quality alarms are active;
- voltage measurements already indicate proximity to operating limits.

The drift monitor is updated from audited scenarios. When new AC outcomes show systematic residual growth, the DMS can inflate intervals, switch to a global fallback radius, disable release, or request recalibration. These governance actions are part of the engineering contribution because they determine when a screen is allowed to affect operations.

7. Algorithmic Decision Logic

Algorithm 1 gives the input-validation and queue-assignment logic. The parameter values are concrete prototype defaults: stale telemetry means a measurement or forecast timestamp older than 5 minutes; near-limit watch means an interval margin below 0.003 p.u.; unseen topology means the topology id is absent from the calibration registry.

```
Algorithm 1: Input validation and queue assignment
Input: scenario x, topology tau, forecast time t_f,
       interval I, registry C, current time t
valid <- true; reasons <- empty
if t - t_f > 5 min:
    valid <- false; add stale_telemetry
if tau not in C:
    valid <- false; add unseen_topology
if x outside calibrated envelope:
    valid <- false; add out_of_envelope
if I intersects voltage limits:
    add interval_limit_intersection
if not valid or interval_limit_intersection:
```

```
    queue <- corrective_audit
else if voltage-limit margin(I) < 0.003 p.u.:
    queue <- watch
else:
    queue <- release
write queue record with reasons and model ids
```

Algorithm 2 handles conservative fallback. Fallback is evaluated after the screen has run because some failure modes are visible only in the interval or residual output.

```
Algorithm 2: Fallback rule evaluation
Input: queue q, protection flags, weather flags,
       residual alarm flag, operator override flag
if protection event or active switching:
    set q <- corrective_audit
if storm mode or data-quality alarm:
    set q <- corrective_audit
if residual alarm is true:
    set q <- corrective_audit
if operator requests escalation:
    set q <- corrective_audit
if operator requests release:
    require supervisor approval and audit note
append fallback reason codes
```

Algorithm 3 defines the rolling drift audit. It uses only scenarios that have subsequent AC labels from routine audit, operator escalation, or corrective study.

```
Algorithm 3: Rolling drift audit
Input: audited outcomes W, calibration quantile q_cal,
       inflation gamma, alarm threshold beta
compute residual scores s_k = |v_ac,k - v_pred,k|
compute rho = mean(s_k > q_cal)
compute coverage and missed counts by family
if rho > beta or a critical family undercovers:
    residual_alarm <- true
if residual_alarm:
    inflate intervals or disable release
if alarm persists for M windows:
    request recalibration and model review
```

8. Implementation Contract

An implementation should expose the following artifacts at every operating cycle:

Artifact	Required fields	Purpose
Scenario record	scenario id, horizon, forecast sources, timestamp	Reconstruct the screened operating state
Network record	feeder id, model version, topology id, voltage limits	Bind the screen to a physical model
Validity record	freshness flags, range checks, topology membership	Decide whether release is allowed
Risk record	bus risk flags, scenario risk score, margin to limits	Rank watch and corrective-audit queues
Queue record	release/watch/audit label, reason code, override flag	Provide auditable DMS action semantics
Audit record	downstream AC result, corrective backend, operator note	Close the feedback loop for drift monitoring

This contract separates the DMS paper from the method paper. The DMS paper is about making a screen operationally accountable; the method paper is about how the interval is estimated.

9. Prototype Demonstration

A Python prototype instantiates the architecture on the archived VoltGuard seed-7 screening outputs. The prototype simulates seven days at 10-minute operating cycles, producing 1008 queue records. Each cycle samples a forecasted scenario, telemetry age, topology-membership status, out-of-envelope forecast flag, operator override flag, and rolling residual alarm flag. The prototype then executes Algorithm 1 and Algorithm 2 and writes an auditable weekly log.

Queue	Records	Violating records	Stale telemetry	Unseen topology	Out-of-envelope	Operator overrides	Record share
corrective audit	846	779	277	24	43	18	0.8393
release	136	0	0	0	0	0	0.1349
watch	26	0	0	0	0	0	0.0258

The prototype is intentionally conservative: 846 of 1008 records route to corrective audit because the simulated week contains many risky scenarios and because stale telemetry, unseen topology, out-of-envelope forecasts, and residual alarms force escalation. The important deployment property is not a large release count; it is that all 136 released records are non-violating in the archived AC labels and every escalation has a reason code. The weekly log also records 18 operator overrides, 277 stale-telemetry events, 24 unseen topology events, 43 out-of-envelope forecasts, and 30 residual alarms.

10. Validation Protocol for Deployment

A deployment validation should not rely only on bus-level prediction metrics. The DMS should verify the following system properties before enabling release:

- every queue record is reproducible from archived inputs and model versions;
- every release has a recorded validity pass and calibration-envelope match;
- every audit-triggered scenario has a reason code;
- sampled released scenarios are periodically checked by AC analysis;
- online residual statistics are logged and tied to recalibration decisions;
- operator overrides are preserved for post-event review.

This validation protocol is intentionally independent of any single learning model. It can be applied to a LinDistFlow baseline, a topology-aware residual screen, or a future OPF-assisted screening module.

11. Discussion

The main risk in DMS screening is not that a model has imperfect offline accuracy; all fast screens have imperfect offline accuracy. The main risk is that the DMS cannot tell when the model is outside its valid operating envelope. VoltGuard-ECM addresses this by making the envelope explicit and by requiring release/watch/audit decisions to carry reason codes and fallback status.

The architecture is intentionally conservative. It does not certify AC feasibility, does not replace OPF or MPC, and does not prove safety under arbitrary nonstationarity. Its value is that it turns voltage-risk screening from an offline score into an auditable DMS object.

The target venue should match this systems-engineering contribution. Energy Conversion and Management can be appropriate if the paper is framed as operational integration for high-DER energy management. Electric Power Systems Research or IEEE Transactions on Power Systems may be a more direct fit if the editorial emphasis is DMS architecture, standards, and operator workflow rather than energy-conversion performance.

Real-world validation remains necessary. The prototype demonstrates the software contract on archived benchmark scenarios, not on a utility DMS. A field pilot should bind the same records to live SCADA/AMI streams, CIM model versions, cybersecurity controls, and operator override logs.

12. Conclusion

VoltGuard-ECM defines a DMS integration architecture for AC-audited voltage-risk screening. Its scientific question is system integration: what contracts, queues, fallback rules, and audit records are required before a calibrated screen can influence operations? By avoiding reuse of the companion method paper's performance results, the manuscript stands as an engineering governance paper rather than a repackaged method evaluation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Code and Data Availability

The experiments use publicly available IEEE test systems implemented through pandapower and a project-local IEEE 69-bus feeder implementation. The local reproducibility package includes the scenario generator, configured evaluation pipeline, conformal calibration code, raw predictions, conformal scores, runtime tables, post-action AC audit outputs, DMS prototype logs, reviewer-requested baseline comparisons, and energy-management value metrics. File-level checksums and reproduction commands are recorded in `experiments/results/reproducibility_manifest.json` and `experiments/results/reproducibility_` The Python implementation, synthetic scenario-generation scripts, trained model artifacts, table-generation scripts, manuscript sources, and release PDFs are publicly archived at `https://github.com/Zhaosiqiang/voltguard-voltage-risk-screening/releases/tag/v1.0.0-subm` and on Zenodo with DOI `10.5281/zenodo.21149702`.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used AI-assisted coding and language tools for drafting support, code scaffolding, consistency checks, and language refinement. The authors reviewed and edited all generated text and code, verified the equations and numerical claims, and take full responsibility for the content of the article.

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